```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score, confusion_matrix, ConfusionMatr

df = pd.read_csv('/content/AttendanceMarks.csv')
df = df.dropna()

X = df[['Attendance', 'ESE']]
y = np.where(df['ESE'] >= df['ESE'].median(), 1, 0)

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

models = {
    'Linear SVM': SVC(kernel='linear'),
    'Non-Linear SVM': SVC(kernel='rbf')
}

fig, axes = plt.subplots(1, 2, figsize=(14, 5))

for ax, (name, model) in zip(axes, models.items()):
    model.fit(X_train_scaled, y_train)
    predictions = model.predict(X_test_scaled)

    print(name)
    print('Accuracy:', accuracy_score(y_test, predictions))
    print('Confusion Matrix:\n', confusion_matrix(y_test, predictions))
    print()

    x_min, x_max = X_train_scaled[:, 0].min() - 1, X_train_scaled[:, 0].max
    y_min, y_max = X_train_scaled[:, 1].min() - 1, X_train_scaled[:, 1].max

    xx, yy = np.meshgrid(
        np.linspace(x_min, x_max, 300),
        np.linspace(y_min, y_max, 300)
    )

    Z = model.predict(np.c_[xx.ravel(), yy.ravel()]).reshape(xx.shape)

    ax.contourf(xx, yy, Z, alpha=0.3, cmap='coolwarm')
    ax.scatter(
        X_train_scaled[:, 0], X_train_scaled[:, 1],
        c=y_train, cmap='coolwarm', edgecolors='black'
    )
```

```
    ax.set_title(name)
    ax.set_xlabel('Attendance (scaled)')
    ax.set_ylabel('ESE (scaled)')

plt.tight_layout()
plt.show()
```

```
Linear SVM
Accuracy: 1.0
Confusion Matrix:
 [[7 0]
 [0 7]]

Non-Linear SVM
Accuracy: 1.0
Confusion Matrix:
 [[7 0]
 [0 7]]
```

